

Algebra 1
Unit 6
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

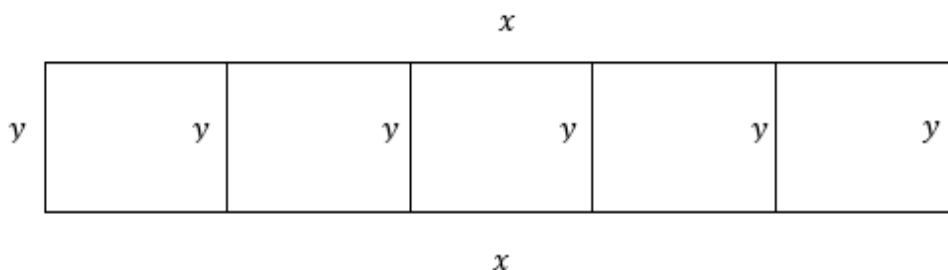
Jaime wants to build a rectangular fence in his yard that will be divided into five equal pens for chickens. Each pen needs to be the same size. He has 36 feet of fencing to build the fence. This scenario can be modeled as a two-variable inequality.

1. Describe what the two variables would represent in the context of Jaime's fence.

x will represent length of the rectangular fence.

y is the sides of the rectangular fence and individual pens.

2. The diagram below shows a possible model of Jaime's fence with sides labeled as x and y .



3. Recall Jaime had 36 feet of fencing to use. Write a two-variable inequality that Jaime could use to find possible dimensions for his fence.

$$2x + 6y \leq 36$$

4. Explain how you chose which inequality symbol to use.

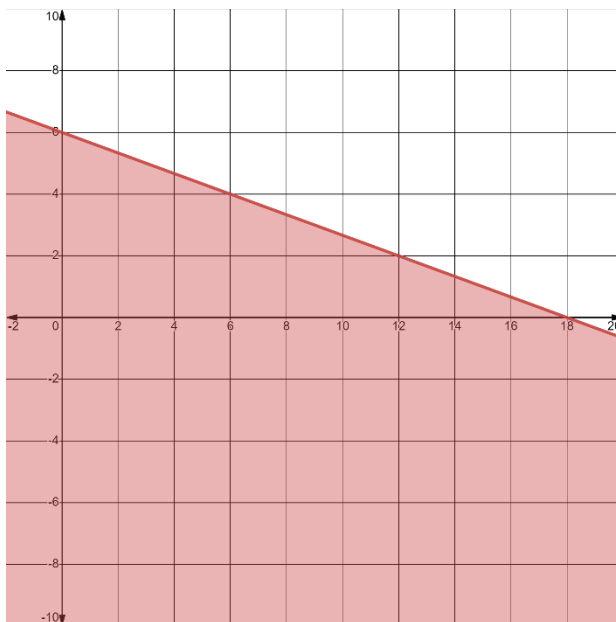
He only has 36 ft. so he can use at most 36 ft., so he needs to use less than or equal to 36.

5. Graph the inequality.

$$\begin{array}{rcl} 2x + 6y & \leq & 36 \\ -2x & & -2x \end{array}$$

$$\frac{6y}{6} \leq \frac{-2x+36}{6}$$

$$y \leq -\frac{1}{3}x + 6$$



Eric was applying for a personal loan to pay for home improvement projects. His bank requires a minimum of \$600 between his checking and savings accounts to be considered for a loan.

6. Define two variables for this situation.

Let x = checking account balance

Let y = savings account balance

7. Write an inequality in terms of x and y that could be used to determine whether Eric meets the minimum requirement for the loan.

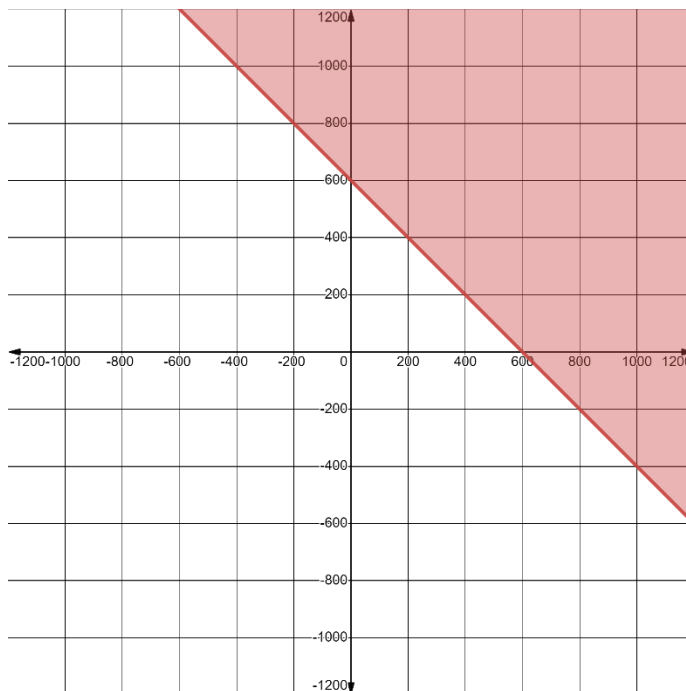
$$x + y \geq 600$$

8. Graph the inequality.

$$x + y \geq 600$$

$$-x \quad -x$$

$$y \geq -x + 600$$



9. Complete the statement.

To be a viable solution in this context, the x and y values must both be rounded to 2 decimal places. Money is rounded to 2 decimal places.

10. Eric overdrew his checking account, which now has a balance of $-\$275.00$. He has $\$957.23$ in his savings account. Does he meet the minimum requirement?

$$\begin{array}{r} 957.23 \\ -275.00 \\ \hline 682.23 \end{array}$$

He has a positive balance of \$682.23 so he does have enough.